Loss: $MSE = \frac{1}{N}\sum_{i=1}^{n}(\hat{y}-y)^2$; Model: $\hat{y} = \omega_1 x_1 + \omega_0$.

Gradients: $\frac{\partial L}{\partial \omega_1} = \frac{2}{N}(\hat{y}-y) \times \frac{\partial L}{\partial \omega}(\omega_1 x_1 + \omega_0) = \frac{2}{N}(\hat{y}-y)x_1$

$\frac{\partial L}{\partial \omega_0} = \frac{2}{N}(\hat{y}-y) \times \frac{\partial L}{\partial \omega_0}(\omega_1 x_1 + \omega_0) = \frac{2}{N}(\hat{y}-y) \times 1$

Update: $\omega_{1new} = \omega_{1old} - \alpha \times \frac{\partial L}{\partial \omega_1}$

$\omega_{0new} = \omega_{0old} - \alpha \times \frac{\partial L}{\partial \omega_0}$

Let $\alpha = 0.1$

#1

$(1,3)\ (2,5)$; $\hat{y}_1 = 0$, $\hat{y}_2 = 0$. error $= \hat{y} - y = -3, -5$;

$Loss = \frac{1}{2}(9+25) = 17$.

Itr	ω_0	ω_1	errors	Loss cal	Loss
1	0	0	-3, -5	$\frac{1}{2}9 + \frac{1}{2}25$	17

#2

$\omega_{0new} = 0 - 0.1 \times \frac{2}{2}(0-3+0-5)$

$= -0.1 \times -8 = +0.8$

$\omega_{1new} = 0 - 0.1 \times \frac{2}{2}(0-3)\times 1 + 0-5 \times 2$

$= -0.1 \times (-3-10) = +1.3$.

$\hat{y}_1 = (1.3)\times 1 + 0.8 = 2.1$ $\hat{y}_2 = (1.3)\times 2 + 0.8 = 3.4$.

$e = (-0.9, -1.6)$

Itr	ω_0	ω_1	errors	Loss cal	Loss
2	0.8	1.3	-0.9, -1.6	$\frac{(-0.9)^2 + (-1.6)^2}{2} = \frac{0.81 + 2.56}{2}$	1.685.